

```
print("Name:KHAN ALFIA SHAMSUL UIN:231A025")
text = "i need to learn new skills before this semester"
```

Name:KHAN ALFIA SHAMSUL UIN:231A025

```
print("Name:KHAN ALFIA SHAMSUL UIN:231A025")
import nltk
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer
import string
```

Name:KHAN ALFIA SHAMSUL UIN:231A025

```
print("Name:KHAN ALFIA SHAMSUL UIN:231A025")
import nltk
nltk.download('punkt_tab')
nltk.download('wordnet')
nltk.download('omw-1.4')
```

Name:KHAN ALFIA SHAMSUL UIN:231A025
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
[nltk_data] Package omw-1.4 is already up-to-date!
True

```
print("Name:KHAN ALFIA SHAMSUL UIN:231A025")
import string
import builtins
print(string.punctuation)
```

Name:KHAN ALFIA SHAMSUL UIN:231A025
! " # \$ % & ' () * + , - . / : ; < = > ? @ [\] ^ _ ` { | } ~

```
print("Name:KHAN ALFIA SHAMSUL UIN:231A025")
import builtins
```

```
text_lower = text.lower()
print(text_lower)
```

Name:KHAN ALFIA SHAMSUL UIN:231A025
i need to learn new skills before this semester

```
print("Name:KHAN ALFIA SHAMSUL UIN:231A025")
import nltk
nltk.download('punkt_tab')
tokens = word_tokenize(text_lower)
print(tokens)
```

Name:KHAN ALFIA SHAMSUL UIN:231A025
['i', 'need', 'to', 'learn', 'new', 'skills', 'before', 'this', 'semester']
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!

```
print("Name:KHAN ALFIA SHAMSUL UIN:231A025")
text_clean = []
for word in tokens:
    if word not in string.punctuation:
        text_clean.append(word)
builtins.print(text_clean)
```

Name:KHAN ALFIA SHAMSUL UIN:231A025
['i', 'need', 'to', 'learn', 'new', 'skills', 'before', 'this', 'semester']

```
print("Name:KHAN ALFIA SHAMSUL UIN:231A025")
lemmatizer = WordNetLemmatizer()
```

```
lemmas = [lemmatizer.lemmatize(tok , pos='v') for tok in text_clean]
builtins.print(lemmas)
```

Name:KHAN ALFIA SHAMSUL UIN:231A025

['i', 'need', 'to', 'learn', 'new', 'skills', 'before', 'this', 'semester']

```
print("Name:KHAN ALFIA SHAMSUL UIN:231A025")
lemmatizer.lemmatize("skills", pos='v')
```

Name:KHAN ALFIA SHAMSUL UIN:231A025

'skills'

```
[(token, lemma) for token, lemma in zip(tokens, lemmas)]
```

```
[('i', 'i'),
 ('need', 'need'),
 ('to', 'to'),
 ('learn', 'learn'),
 ('new', 'new'),
 ('skills', 'skills'),
 ('before', 'before'),
 ('this', 'this'),
 ('semester', 'semester')]
```

```
morphological_info = {}
for token, lema in zip(tokens, lemmas):
    if token != lema:
        if token.endswith('ing'):
            analysis = "Inflectional: Present participle/continous form of '{}'.format(lema)
        elif token.endswith("er") or token.endswith("ation"):
            analysis = "Derivational: '{}' + -er -> person who does the action".format(lema)
        else:
            analysis = "Inflectional or irregular form of '{}'.format(lema)
    else:
        analysis = "Base/root form"
    morphological_info[token]=analysis
```

```
print(morphological_info)
```

```
{'i': 'Base/root form', 'need': 'Base/root form', 'to': 'Base/root form', 'learn': 'Base/root form', 'new': 'Base/root form', 'skills': 'Base/root form', 'before': 'Base/root form', 'this': 'Base/root form', 'semester': 'Base/root form'}
```